

## Scalability & Future Plan

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|----------------------|-----------------------------------------------------|
| <b>Date</b>          | 15 March 2024                                       |
| <b>Team ID</b>       | SWTID-2026-5339                                     |
| <b>Project Name</b>  | EduGenie – Google Gemini Powered Learning Assistant |
| <b>Maximum Marks</b> | 1 Mark                                              |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                                                                                    | Impact                                                           | Priority to Address<br>(High / Medium / Low) |
|------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------|
| 1    | The application currently supports only local deployment using Uvicorn and is not hosted on a cloud platform. | Limited accessibility for multiple users.                        | High                                         |
| 2    | User data, learning history, and quiz progress are not stored because no database is integrated.              | No personalized progress tracking or history management.         | High                                         |
| 3    | The project does not include user authentication and role-based access control.                               | Anyone can access the application without personalized accounts. | Medium                                       |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                                                              | Proposed Upgrade / Solution                                                                                                    |
|------|--------------------|--------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports a limited number of users in a local environment.                                 | Deploy the application on cloud platforms such as AWS, Azure, or Google Cloud with load balancing to support concurrent users. |
| 2    | Data Storage       | No database is used for storing user information or learning history.                      | Integrate PostgreSQL or MongoDB to store user profiles, quiz history, summaries, and learning progress.                        |
| 3    | Performance        | AI responses depend on external Google Gemini API calls, which may increase response time. | Implement caching, asynchronous API calls, and optimized prompt handling to improve response speed.                            |
| 4    | Security           | API key is stored locally and no user authentication is implemented.                       | Add secure API key management, JWT-based authentication, HTTPS, and role-based access control.                                 |

### Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|-------|-------------------------------|-----------------|-----------------|
|-------|-------------------------------|-----------------|-----------------|

|         |                                                                                                     |            |                                                                                                               |
|---------|-----------------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------|
| Phase 2 | User authentication, profile management, and learning progress tracking                             | 2–3 Months | Provide personalized learning experiences and secure user accounts.                                           |
| Phase 3 | Database integration and analytics dashboard                                                        | 3–4 Months | Store learning history, quiz scores, and provide progress analytics for students.                             |
| Phase 4 | Mobile application, multilingual support, voice-based interaction, and AI-powered adaptive learning | 4–6 Months | Improve accessibility, support diverse learners, and provide an intelligent personalized learning experience. |
| Phase 1 | Cloud deployment with a production-ready FastAPI server                                             | 1–2 Months | Enable online access for multiple users and improve application availability.                                 |

### Future Vision:

The long-term vision of EduGenie – Google Gemini Powered Learning Assistant is to evolve into a comprehensive AI-powered educational platform that supports personalized learning, adaptive assessments, multilingual content generation, voice-enabled interaction, and cloud-based accessibility. By integrating advanced AI capabilities and scalable infrastructure, the system aims to provide students and educators with an intelligent, secure, and highly interactive learning environment.